

DSAT (Jan - April 2025)

Assignment 2

Instructions:

1. Each question has only one right choice.
2. Each question carries one mark.
3. Google form to fill the answers will be shared with you later.
4. Deadline for submission is 15 April midnight.

1. What is the runtime of the below code?

Fun(n):

if $n \leq 1$

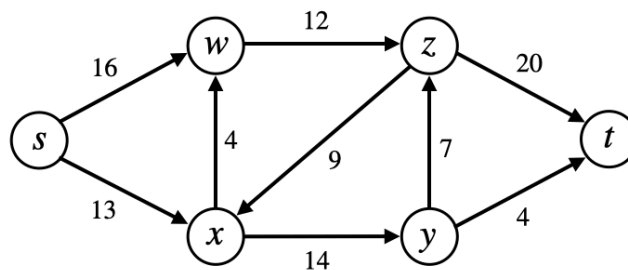
return n

else

return **Fun**($n - 1$) + **Fun**($n - 2$)

- a) $O(n^2)$
- b) $O(n)$
- c) $O(2^n)$ ✓
- d) $O(n \log n)$

2. In the below graph, what is the correct order of vertices dequeued from the priority queue if we want to find the shortest paths from the source vertex s using Dijkstra's algorithm?



- a) $s, x, y, w, z, t.$
- b) $s, x, w, z, y, t.$
- c) $s, x, w, y, z, t.$ ✓
- d) $s, x, w, y, t, z.$

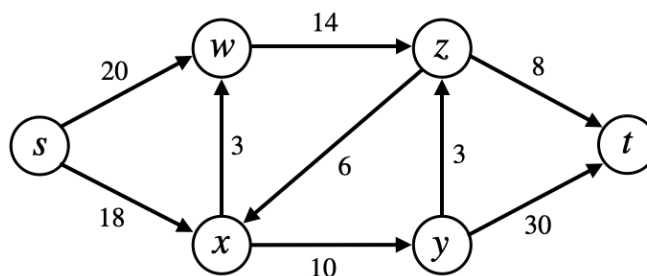
3. Which of the following statement is neither known to be true nor known to be false?

- a) Some problems in **NP** cannot be solved in polynomial time.
- b) A polynomial-time algorithm for an **NPC** problem implies polynomial-time algorithm for all the **NP** problems.
- c) Every problem in **P** is also in **NP**.
- d) None of the above.

4. Which of the following statements is true with respect to Dijkstra's algorithm?

- a) Dijkstra's algorithm always computes correct shortest paths even when edges have negative weights.
- b) Dijkstra's algorithm may compute correct shortest paths when edges have negative weights.
- c) Dijkstra's algorithm will never compute correct shortest paths when edges have negative weights.
- d) Dijkstra's algorithm does not work in undirected weighted graphs.

5. Find the value of the maximum flow in the below flow network with s as the source and t as the sink?



- a) 20.
- b) 18.
- c) 17.
- d) 19.

6. What is the length of the longest common subsequence of the following two sequences of numbers?

$S1 = \langle 1, 2, 1, 2, 4, 1, 5, 4, 9, 8, 6, 3, 1, 3, 1, 3, 7, 5, 2, 3, 2, 6, 2 \rangle$

$S2 = \langle 1, 3, 2, 4, 6, 2, 2, 8, 9, 7, 6, 5, 2, 4, 5, 2, 7, 2, 6, 5, 5, 2, 1 \rangle$

- a) 15.
- b) 12.
- c) 8.
- d) 10.

7. **Longest Increase Subsequence** problem asks to compute the length of a longest subsequence of the given sequence of integers such that the elements of that subsequence are sorted in ascending order.

For instance, in sequence $\langle 0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15 \rangle$, $\langle 0, 2, 6, 9, 11, 15 \rangle$ is one of the longest increasing subsequence.

Below is the pseudocode for **LIS** problem with line 7 missing.

```
1. LIS( $S$ ):  
2.    $n = \text{sizeof}(S)$   
3.    $dp[1 : n] = \{1, 1, \dots, 1\}, ans = 1$   
4.   for  $i = 2$  to  $n$   
5.      $k = 0$   
6.     for  $j = 1$  to  $i - 1$   
7.  
8.        $k = \text{Max}(k, dp[j])$   
9.      $dp[i] = k + 1$   
10.     $ans = \text{Max}(ans, dp[i])$   
11.  return  $ans$ 
```

Which of the following should be line 7?

- a) **if** $S[j] \neq S[i]$.
- b) **if** $S[j] > S[i]$.
- ☒ c) **if** $S[j] < S[i]$.
- d) None of the above.

8. Which of the following statements are false?

- ☒ a) Ford-Fulkerson method always terminates with the max flow value.
- b) Ford-Fulkerson method always returns an integer-valued flow, when capacities are integers.
- c) If there is no augmenting path in a residual network, then the corresponding flow is a max flow.
- d) If there is an augmenting path in a residual network, then the corresponding flow is not a max flow.